EpiKG: End-to-End Epistemic Preservation in Clinical Knowledge Graphs for Assertion-Aware Retrieval-Augmented Generation

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Abstract

Clinical NLP detects assertion status—negation, uncertainty, family history—yet to our knowledge, prior KG-RAG systems do not propagate these labels into retrieval-augmented generation. We call this the *epistemic propagation gap*. EpiKG closes it with a 7-value assertion schema on every edge, tri-temporal modeling, and intent-aware retrieval that dispatches question-type-specific graph traversals. On ClinicalBench ($n = 400$, MIMIC-IV, Claude Opus 4.6), intent-aware KG-RAG achieves 69% ($+47$ pp vs. LLM alone), improving all 9 categories. An ablation decomposition isolates two additive components: structured retrieval contributes $+28$ pp, epistemic annotations a further $+19$ pp—a roughly 60/40 split. Change detection exemplifies the distinction: retrieval without assertions reaches 17%, but adding assertion metadata lifts it to 100%. Brute-force long context collapses to 0% on historical and sequence despite a 200K token window. Cross-model validation on three models shows KG-RAG benefit is consistent with an inverse relationship to parametric medical knowledge, suggesting a knowledge-compensator effect. A blinded physician pilot ($n = 30$) independently confirms the effect: KG-RAG answers were rated correct in 81% vs. 29% for LLM-alone, safe in 88% vs. 50%, and helpful in 81% vs. 29%, while revealing that 40% of auto-generated gold standards require revision. The category$\times$condition interaction—not the aggregate—is the right evaluation lens for clinical KG-RAG.

1 Introduction

A physician documents "patient denies chest pain, sister had MI at 45, will consider statin if lipids remain elevated"—a single sentence encoding four epistemic states: a negated symptom, a family-history event, a hypothetical action, and an implicit present condition. Clinical NLP can detect these assertions accurately [1, 2], yet when a RAG system ingests the same note, assertion context is flattened: negation is lost, family attribution collapses onto the patient, and downstream reasoning may conclude the patient *has* chest pain and *had* an MI. We call this the *epistemic propagation gap*—the systematic loss of assertion status as clinical information moves from extraction through storage to retrieval.

The gap persists because, to our knowledge, prior KG-RAG systems do not propagate assertion labels beyond the extraction stage into graph-augmented retrieval. Clinical NLP pipelines (cTAKES, John Snow Labs) detect assertions, but do not carry them as edge attributes into knowledge graph construction or retrieval. OMOP (Observational Medical Outcomes Partnership) excludes negated conditions from `CONDITION_OCCURRENCE` [3]; FHIR (Fast Healthcare Interoperability Resources) provides `verificationStatus` for `Condition` resources only, covering a fraction of the clinical data model. Graph-augmented RAG systems—GraphRAG [4], GFM-RAG [5], KARE [6], Medical-Graph-RAG [7]—build KGs that discard the metadata distinguishing "patient has diabetes" from "rule out diabetes." Two KG surveys [8, 9] do not identify assertion preservation as a concern. A parallel *temporal integration gap* exists: clinical events span three time dimensions (valid time, transaction time, NLP-asserted temporality), yet existing systems model at most two [10, 11]. Allen's interval algebra [12] has been formalized as annotation ontologies [13] and modal operators [14], but never as first-class KG edge attributes that participate in retrieval.

We present EpiKG, a clinical KG system that closes both gaps by preserving epistemic status end-to-end from NLP extraction through OMOP mapping, fact construction, KG materialization, and graph-augmented retrieval. Our contributions:

1. **Epistemic preservation invariant.** We formalize the requirement that assertion status be recoverable at every downstream stage, give an information-theoretic bound on the loss from assertion-blind pipelines (Section 3.5, Appendix B), and realize the invariant with a seven-value schema extending i2b2 [1], carried from extraction through OMOP mapping, fact construction, KG materialization, and RAG serialization.
2. **Tri-temporal KG with Allen's interval algebra.** Edges carry valid time, transaction time, and NLP-asserted temporality alongside nine temporal relations (seven from Allen's algebra [12] plus CONCURRENT and UNKNOWN) as first-class attributes, enabling temporal reasoning over longitudinal records.
3. **Ablation with decomposition and bookend baselines.** We evaluate retrieval conditions—from C7 (deterministic KG, no LLM; 3.8%) through C4g (intent-aware KG-RAG; 69.0%, $+47$ pp vs. LLM alone)—on ClinicalBench and SliceBench. A new ablation step C3 (KG-RAG without assertions; 50.0%) decomposes the gain: structured retrieval contributes $+28$ pp, epistemic annotations $+19$ pp. Bookend baselines C6 (brute-force long context; 39.0%) and C7 bracket the design space. Intent-aware KG-RAG improves all 9 categories; neither more context nor structured data alone can substitute for LLM + epistemic KG-RAG.
4. **ClinicalBench, SliceBench, and cross-model validation.** Two benchmarks (400 and 144 questions) across 9 epistemic categories target the propagation gap, with hard longitudinal questions (change, cross-admission, temporal) as the primary endpoint. Cross-model validation on MedGemma 27B and GPT-OSS 20B shows KG-RAG benefit consistent with an inverse relationship to parametric medical knowledge; a physician adjudication pilot ($n = 30$, Section 5.5) validates the automated evaluator.

The central research question is: *under what conditions does structured epistemic context help, hurt, or break even compared to unstructured retrieval or no retrieval at all?* Empirically, we observe: intent-aware KG-RAG improves every category, with the largest gains on longitudinal reasoning that requires cross-admission synthesis. An ablation decomposition shows that structured retrieval provides the majority of the gain ($+28$ pp); epistemic annotations are critical for change detection, where assertion metadata alone accounts for an $+83$ pp jump. Brute-force long context fails on historical and sequence categories despite a 200K token window. Cross-model validation shows the benefit is consistent with an inverse relationship to parametric medical knowledge, suggesting a knowledge-compensator effect. The magnitude and direction of the effect are both category- and model-specific, producing interactions that aggregate accuracy alone obscures (Section 5).

2 Related Work

We organize prior work along four axes and synthesize the gaps that EpiKG addresses (Table 8, Appendix).

Medical RAG systems. Graph-augmented retrieval has emerged as a leading paradigm for medical QA. GraphRAG [4] introduced community-based summarization; GFM-RAG [5] trained an 8M-parameter graph foundation model across 60 KGs (14M+ triples); KARE [6] adapted community retrieval to clinical decision support; DoctorRAG [15] emulates physician reasoning; MedRAG [16] constructs hierarchical diagnostic KGs; Medical-Graph-RAG [7] links documents via triple graphs. Concurrent work addresses EHR-specific retrieval: EHR-RAG [17] evaluates retrieval strategies over structured EHR tables, and MediGRAF [18] builds medical KGs via multi-agent extraction. Yet no published medical RAG system models assertion status or temporal context; GFM-RAG and KARE build population-level graphs rather than patient-level graphs from clinical text, and a recent survey [19] confirms no system incorporates epistemic metadata into the retrieval graph.

Clinical KG construction. Multi-LLM KG-RAG [20] uses multi-agent LLM prompting with schema-constrained extraction for oncology, modeling uncertainty via entropy but not tracking as-

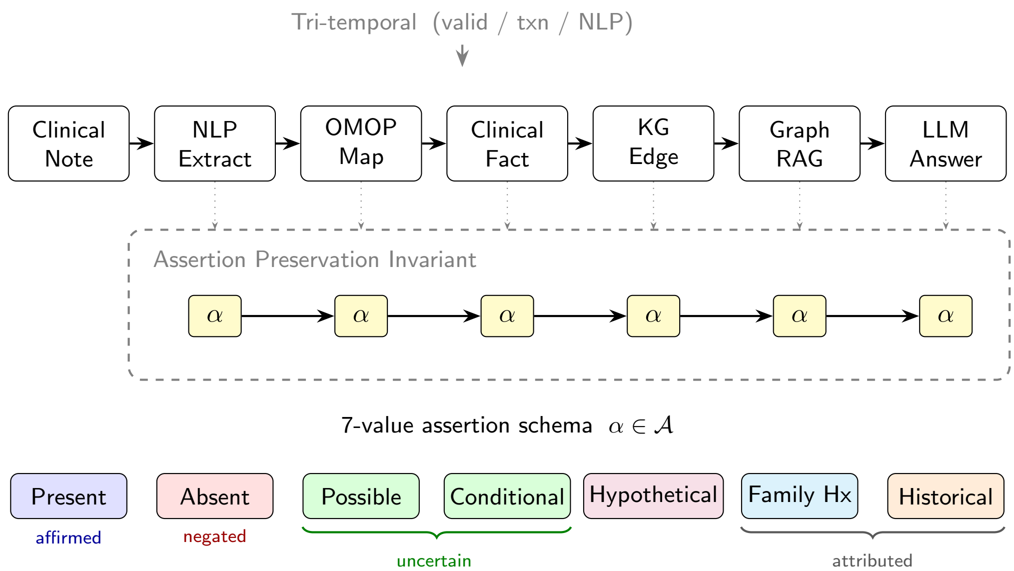

Figure 1: The EpiKG pipeline. The assertion label α (highlighted) is preserved as a first-class attribute from NLP extraction through OMOP mapping, fact construction, KG materialization, and assertion-aware GraphRAG.

sertion classes on edges. AutoRD [21] applies GPT-4 to rare disease literature; RECAP-KG [22] handles messy GP notes; FHIR-Ontop-OMOP [23] generates virtual KGs from OMOP data. All share a common limitation: assertion status is not propagated into the final graph.

Assertion detection. NegEx [24] introduced trigger-based negation detection; ConText [25] extended it with temporality and experiencer. The i2b2/VA shared task [1] formalized a six-class taxonomy; Gul et al. [2] fine-tuned LLMs to 0.962 accuracy. All treat assertion detection as a terminal annotation task: labels are not carried into knowledge graphs or retrieval systems.

Assertion detection is necessary but not sufficient: labels must also be *temporally situated*—the same condition may be present in one encounter and absent in the next.

Temporal knowledge graphs. MedTKG [10] constructs temporal KGs with time-stamped snapshots (event time only); Graphiti [11] implements bitemporal edges [26] but lacks clinical ontology alignment; TEO [13] and TCL [14] formalize Allen's relations [12] as annotation ontologies or modal operators rather than KG edge attributes. Uncertainty quantification in biomedical KGs [27] has not been connected to assertion taxonomies. Two structural gaps emerge: an *epistemic propagation gap* (assertion-detecting systems do not persist labels into KGs) and a *temporal integration gap* (temporal formalisms operate as annotation layers, not as KG edge attributes participating in retrieval). EpiKG closes both by carrying assertion and temporal metadata as first-class properties through every pipeline stage.

3 System Design

3.1 Overview

EpiKG transforms unstructured clinical narratives into an assertion-aware, temporally grounded knowledge graph exposed through graph-augmented retrieval. The pipeline comprises six stages: (1) document ingestion, (2) NLP extraction, (3) OMOP concept normalization [3], (4) clinical fact construction with epistemic metadata, (5) KG materialization with shared concept nodes, and (6) assertion-aware retrieval. The key invariant is that *epistemic assertion status* is a first-class attribute at every stage, not an NLP annotation discarded after extraction (Figure 1). Clinical concepts are stored once as shared global nodes; patient-specific semantics (assertion status, temporality, confidence) reside on edges, enabling cohort queries via edge scans.

3.2 Epistemic Assertion Schema

Clinical notes abound with qualified statements—"*no evidence of pneumonia*," "*possible CHF*," "*mother had breast cancer*"—yet standard representations discard these qualifiers. OMOP excludes negated conditions entirely [3]; FHIR limits assertion metadata to conditions. EpiKG defines a seven-value assertion taxonomy:

$$\alpha \in \{\text{Pres., Abs., Poss., Cond., Hypo., Fam.Hx., Hist.}\} \tag{1}$$

This extends the i2b2 (Informatics for Integrating Biology and the Bedside) six-class taxonomy [1] by separating HISTORICAL from FAMILY_HISTORY, a clinically important distinction that prior work conflates (definitions in Appendix Q). A rule-based classifier (101 scope-aware trigger patterns) assigns α with an associated confidence score; we chose rule-based classification for deterministic behavior and interpretability, though fine-tuned LLM classifiers [2] could be substituted. Intrinsic classifier evaluation is deferred; the end-to-end ClinicalBench ablation (C3→C4g) serves as a functional test of assertion propagation. The label is carried through every stage: from `ExtractedMention` through ensemble merging, `ClinicalFact` records, KG edge properties, and retrieval scoring. To our knowledge, no prior system [2, 28] maintains this end-to-end invariant.

3.3 Tri-Temporal Knowledge Graph Model

Clinical events unfold across multiple time dimensions that prior systems model incompletely [10, 11]. Bi-temporal databases [26] distinguish valid time from transaction time; EpiKG adds a third dimension (NLP-asserted temporality) and stores all three on every KG edge:

Definition 1 (Tri-Temporal Edge)**.** *An edge $e = (s, p, o, \alpha, \boldsymbol{\tau}_v, \boldsymbol{\tau}_t, \tau_a, r, c)$ where:*

- *s, o are source and target nodes; p is the predicate (one of 24 edge types);*
- *$\alpha \in \mathcal{A}$ is the assertion label (Eq. 1);*
- *$\boldsymbol{\tau}_v = (\textit{event\_date}, \textit{valid\_from}, \textit{valid\_to})$ is the **valid time** interval—when the relationship held in the real world;*
- *$\boldsymbol{\tau}_t = (\textit{recorded\_at}, \textit{doc\_date}, \textit{created\_at})$ is the **transaction time**—when it was recorded;*
- *$\tau_a \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ is the **NLP-asserted temporality**;*
- *$r \in \mathcal{R}$ is an Allen interval algebra relation [12]; $c \in [0, 1]$ is temporal confidence.*

$\mathcal{R}$ comprises seven of Allen's 13 canonical relations [12] (merging symmetric pairs like Before/Meets) plus CONCURRENT and UNKNOWN (nine total; full mapping in Appendix R). Unlike TEO [13] and TCL [14], which use Allen's relations as annotation-ontology classes or modal operators, EpiKG stores r and c directly on materialized edges, enabling efficient temporal filtering during traversal.

3.4 Assertion-Aware Graph-Augmented Retrieval

Given a clinical question q and patient π, the retrieval pipeline: (1) extracts concepts from q via NLP + OMOP enrichment; (2) traverses 2–3 hops via bidirectional breadth-first search (BFS) over patient KG edges and OMOP vocabulary relationships (20M+ edges) via PostgreSQL common table expressions (CTEs) (Appendix K), pruning edges with $c < 0.3$; (3) groups edges into four temporal views (event timeline, current state, historical, conflicts); (4) retrieves matching guideline sections; (5) scores edges by:

$$\text{score}(e) = c(e) + \mathbb{1}[\text{type}(e) \in Q_{\text{rel}}] \cdot 0.2 + \mathbb{1}[\tau_a(e) = \text{CURRENT}] \cdot 0.1 \tag{2}$$

where the bonuses for question-relevant edge types (0.2) and current temporality (0.1) were set by manual tuning on a development set of 20 questions. The surviving subgraph is serialized into structured text preserving assertion labels (e.g., "ABSENT: pneumonia"); and (6) composes graph evidence, temporal context, guidelines, and source documents into a single prompt.

3.5 Formal Epistemic Preservation

We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the ontological principle that aboutness must be preserved across transformations [29]. The *epistemic state* of a clinical mention m is the tuple $e(m) = (c, \alpha, \xi, \tau)$: OMOP concept, 7-value assertion

Table 1: ClinicalBench conditions. C1–C4g form the ablation ladder; C6 and C7 are bookend baselines.

ID	Condition	Retrieval	Assertion	Temporal
C1	LLM Alone	None	None	None
C2	+ Vanilla RAG	TF-IDF doc chunks	None	None
C3	+ KG-RAG (no assertions)	Graph + Doc	None	None
C4	+ Epistemic KG-RAG	Graph + Doc	Full (7-class)	Tri-temporal
C4g	+ Intent-Aware KG-RAG	Graph + Doc (type-specific)	Full (7-class)	Tri-temporal
C6	Long Context	All notes	None	None
C7	Deterministic KG	KG lookup (no LLM)	Full	Tri-temporal

label, experiencer ({PATIENT, FAMILY}), and temporality ({CURRENT, PAST, FUTURE}); see Table 7 (Appendix) for notation summary. A pipeline *epistemically preserves* m if the output epistemic state equals the extraction state at every stage. We formalize (Appendix B) that an assertion-blind pipeline collapses the assertion distribution to a degenerate point mass, incurring entropy loss $\Delta H = H(A_c^\pi) \geq 0$ [30], and that assertion-faithful accuracy is bounded by $1 - f_{np}(c)$, where f_{np} is the fraction of non-present assertions. The empirical consequences are measured via category-stratified accuracy in Section 5.

3.6 Intent-Aware Retrieval (C4g)

The multi-hop BFS described above treats all questions uniformly, yet different clinical question types require fundamentally different graph operations. A *change* question requires cross-admission set differencing; a *current-state* question needs filtering to the most recent valid edges; a *historical* question must recover resolved conditions.

Intent classifier. A lightweight rule-based classifier maps each question to one of four intents—CHANGE, CURRENT_STATE, HISTORICAL, or DEFAULT—using question metadata with keyword-pattern fallback (Algorithm 1, Appendix S).

Change retrieval. For $\iota =$ CHANGE, edges are partitioned by admission (`hadm_id`). The module computes set differences across admission pairs (k, k')—concepts *added* ($\mathcal{C}_{k'} \setminus \mathcal{C}_k$), *removed* ($\mathcal{C}_k \setminus \mathcal{C}_{k'}$), and *continued* ($\mathcal{C}_k \cap \mathcal{C}_{k'}$)—and returns structured evidence.

Current-state and historical retrieval. For CURRENT_STATE, the module filters to edges with $\tau_a =$ CURRENT or open-ended validity, deduplicates by concept, and emits "NOT FOUND" for missing concepts. For HISTORICAL, it selects edges with $\tau_a =$ PAST and augments with admission-based inference (concepts in earlier admissions but absent from the latest are labeled "resolved"). Each intent triggers a type-specific prompt template that structures evidence according to question semantics (Appendix U).

4 Benchmark Design

Existing clinical QA benchmarks evaluate factual recall (MedQA [31]) or agent task completion [32] but do not test handling of epistemic qualifiers or reasoning over longitudinal records of varying complexity. We introduce two complementary benchmarks built on MIMIC-IV [33].

4.1 ClinicalBench: Assertion-Sensitive Clinical QA

ClinicalBench comprises 400 gold-standard questions over 43 MIMIC-IV patients in two tasks: **Task A** (200 questions, negation-aware retrieval) and **Task B** (200 questions, temporal reasoning). Questions span 9 categories: *negation*, *conditional*, *uncertainty*, *family_history*, *sequence*, *current_state*, *duration*, *historical*, and *change*. Gold-standard answers were created by a single physician (board-certified emergency medicine) through manual chart review against source MIMIC-IV documents. No inter-annotator agreement was computed for gold-standard creation; the physician adjudication pilot (Section 4.4) addresses this limitation. Four ablation conditions progressively add system components (Table 1). Two bookend baselines—C6 (all notes concatenated) and C7 (deterministic KG lookup, no LLM)—bracket the design space.

Endpoints. The **pre-registered primary endpoint** is the hard longitudinal subset (*change* + *current_state* + *historical*, $n = 130$): C4g vs. C1 aggregate accuracy. The **secondary endpoint** is full

400-question ClinicalBench, C4g vs. C1. **Diagnostic endpoints** are per-category C1→C4g accuracy deltas and the C3→C4g transition isolating epistemic annotation lift.

What each transition isolates. C1→C2 isolates the *document retrieval effect*: C2 adds TF-IDF chunk retrieval over clinical notes, analogous to existing medical RAG systems (DoctorRAG, MedRAG). C2→C3 isolates the *graph structure effect*: both retrieve documents, but C3 replaces TF-IDF with graph traversal—testing whether structured retrieval outperforms vanilla retrieval. C3→C4g isolates the *epistemic annotation effect*: both retrieve from the same graph, but C4g carries full 7-class assertions, tri-temporal metadata, and intent-aware routing. C4g vs. C6 isolates the *structured-vs-unstructured retrieval effect* on the same questions.

4.2 SliceBench: Complexity-Stratified Longitudinal QA

SliceBench tests the hypothesis that KG-augmented retrieval provides increasing benefit as patient record complexity grows. We select 6 MIMIC-IV patients in three tiers: **Tier A** (2 patients, 1–2 encounters), **Tier B** (2 patients, 5–10), and **Tier C** (2 patients, 15+). Each patient receives 24 questions (including hard longitudinal categories: cross-encounter medication timelines, problem list reconciliation, causal chain tracing), yielding 144 total. Five conditions (B0–B4) form a monotone context progression from parametric-only to full system (Appendix J). The critical comparison is B2→B3: both have the same documents, but B3 adds structured KG context with assertion metadata and temporal relations.

4.3 Evaluation Protocol

ClinicalBench uses a *deterministic keyword evaluator* (v2) that performs exact word-boundary matching of gold-standard assertion and temporal keywords against model answers (keyword lists in Appendix P). The v2 evaluator includes an *abstention detection gate*: when a model responds that information is unavailable in the notes (e.g., "cannot determine," "not mentioned") rather than making a clinical claim, the answer is scored as incorrect regardless of keyword matches. This prevents false positives where refusal language incidentally matches gold-standard patterns. Sequence and change categories additionally require domain-specific keywords (ordering terms and change terms, respectively). This evaluator is fully reproducible (no stochastic LLM judge) and is used for all ClinicalBench results. The primary answering model is Claude Opus 4.6; cross-model validation uses MedGemma 27B and GPT-OSS 20B under the same evaluator.

SliceBench uses an *LLM-as-judge* protocol with separated answering and judging models to prevent self-evaluation bias [34]: Claude Sonnet 4.5 generates answers and Claude Opus 4.6 judges correctness.

Statistical reporting. We report BCa bootstrap 95% CIs ($n = 2{,}000$, seed 42) [35] on both benchmarks. Question-level resampling is the primary analysis; patient-level cluster bootstrap (resampling the 43 patients with replacement, including all their questions) is reported as a sensitivity analysis to account for within-patient correlation (Appendix Table 16). McNemar's test [36] provides a complementary non-parametric paired comparison. ClinicalBench CIs appear on primary deltas (overall C4g$-$C1 and hard longitudinal subset); SliceBench CIs appear on all conditions (Table 5). A safety score with asymmetric weighting ($w = 2.0$ for false-positive assertion errors) captures clinical risk (Appendix D).

4.4 Physician Validation Protocol

To validate automated scoring, a board-certified physician (A.S.) reviewed a blinded, stratified pilot sample ($n = 30$ ClinicalBench items across all 9 categories and both conditions, C1 vs. C4g). The physician-rated C4g–C1 delta (+52 pp) exceeded the auto-evaluator delta (+47 pp), indicating the headline effect is conservative (Section 5.5). A second independent reviewer is planned; protocol details are in Appendix O.

5 Results

We evaluate EpiKG through two benchmarks: ClinicalBench tests assertion-sensitive and longitudinal reasoning, while SliceBench measures longitudinal QA across patient complexity tiers.

Table 2: ClinicalBench primary results (400 questions, Claude Opus 4.6, keyword evaluator v2 with abstention detection). Ablation ladder (C1–C4g) progressively adds components; C4 isolates assertion metadata from intent routing; C6 and C7 are bookend baselines. Sig: * denotes BCa CI excluding zero. Best in **bold**.

	Condition	Accuracy	Δ vs C1
C1	LLM Alone	21.8%	—
C2	+ Vanilla RAG	52.2%	+30.5 pp *
C3	+ KG-RAG (no assertions)	50.0%	+28.2 pp *
C4	+ Assertions (no routing)	46.8%	+25.0 pp *
C4g	+ Intent-Aware KG-RAG	**69.0%**	+47.2 pp *
C6	Long Context (all notes)	39.0%	+17.2 pp
C7	Deterministic KG (no LLM)	3.8%	−18.0 pp

ClinicalBench uses a deterministic keyword evaluator for reproducibility; SliceBench uses LLM-as-judge evaluation (Section 4.3). Claude Opus 4.6 is the primary answering model; MedGemma 27B and GPT-OSS 20B[1] provide cross-model validation. Both benchmarks report bias-corrected and accelerated (BCa) bootstrap 95% CIs (n = 2,000, seed 42) [35], resampled at the question level.

5.1 ClinicalBench: Primary Ablation

Intent-aware KG-RAG (C4g) reaches 69.0% (+47.2 pp vs. C1; 95% BCa CI: [+40.9 pp, +52.8 pp]; patient-level cluster bootstrap: [+40.2 pp, +52.5 pp]; McNemar $\chi^2 = 157.1$, $p < 10^{-6}$). Without retrieval, the model mostly abstains or produces vague responses that fail keyword matching; with structured retrieval, it produces substantive correct answers. The bookend baselines bracket the design space. C6 (all notes concatenated) scores 39.0%, outperforming C1 (+17.2 pp) because it provides clinical text. However, C6 fails on historical (0%) and sequence (0%), consistent with the "lost in the middle" phenomenon when context exceeds what the model can attend to. C7 (deterministic KG, no LLM) achieves 3.8%—only negation detection works (13.6%) via stored assertion metadata—confirming that an LLM is essential for reasoning over graph evidence.

Ablation decomposition: retrieval vs. assertions vs. intent routing. C2 (vanilla TF-IDF document retrieval) and C3 (graph-augmented retrieval without assertions) score similarly overall—52.2% and 50.0%, respectively—demonstrating that graph structure alone does not improve aggregate accuracy over simple document retrieval. However, per-category patterns diverge: C3 excels on current state (54% vs. 32%) where graph structure helps identify active conditions, while C2 outperforms C3 on change (43% vs. 17%) where raw text explicitly mentions medication changes.

The new C4 condition isolates assertion metadata from intent routing: C4 adds epistemic annotations (negation, uncertainty, temporality) to graph edges but uses uniform BFS traversal without question-type-specific routing. C4 scores 46.8%—*below* C3 (50.0%)—with the C3→C4 delta of −3.2 pp not reaching significance (95% BCa CI: [-10.0 pp, +3.2 pp]; McNemar $p = 0.37$). This reveals that assertion metadata without intent routing actually degrades performance: the additional edge properties increase evidence verbosity and confuse uniform BFS traversal. The entire C3→C4g gain (+19.0 pp; CI: [+13.8 pp, +24.0 pp]) is attributable to intent-aware routing (C4→C4g: +22.2 pp; CI: [+16.0 pp, +28.0 pp]; McNemar $\chi^2 = 41.4$, $p < 10^{-10}$).

Per-category decomposition clarifies the mechanism: assertions without routing help on assertion-sensitive categories (negation: +24.5 pp, uncertainty: +15.0 pp) but *hurt* on temporal categories (current state: −24.0 pp, historical: −30.0 pp, sequence: −47.5 pp). Intent routing reverses these losses and amplifies the gains—most dramatically on change (+90.0 pp), sequence (+55.0 pp), and historical (+48.0 pp). The finding that assertions are necessary but not sufficient—and that their value is unlocked specifically by routing—validates the two-component architecture.

C2 serves as the closest analog to existing medical RAG systems (DoctorRAG, MedRAG), which use document retrieval without assertion-aware graph structure; the +16.8 pp C2→C4g gap quantifies the value of epistemic KG-RAG over vanilla approaches.

[1] GPT-OSS 20B is an open-source general-purpose model run locally via Ollama (4-bit quantized). We use the label GPT-OSS to avoid implying endorsement; the reproducibility package includes model identifiers and raw predictions.

Table 3: ClinicalBench per-category accuracy (%) across conditions (Claude Opus 4.6, evaluator v2). C4 adds assertion metadata to C3 without intent routing; C4g adds intent-aware routing. Best per row in **bold**.

Category	n	C1	C2	C3	C4	C4g	C4g−C1
Negation	110	45.5	70.9	65.5	90.0	**80.9**	↑ 35.5 pp
Conditional	20	0.0	35.0	30.0	**50.0**	45.0	↑ 45.0 pp
Uncertainty	40	12.5	35.0	37.5	52.5	**50.0**	↑ 37.5 pp
Family hist.	30	3.3	43.3	43.3	50.0	**56.7**	↑ 53.3 pp
Sequence	40	10.0	60.0	57.5	10.0	**65.0**	↑ 55.0 pp
Current st.	50	26.0	32.0	54.0	30.0	**70.0**	↑ 44.0 pp
Duration	30	30.0	66.7	60.0	46.7	**66.7**	↑ 36.7 pp
Historical	50	6.0	48.0	42.0	12.0	**60.0**	↑ 54.0 pp
Change	30	6.7	43.3	16.7	10.0	**100.0**	↑ 93.3 pp
Overall	**400**	21.8	52.2	50.0	46.8	**69.0**	↑ 47.2 pp

Table 4: ClinicalBench cross-model results (400 questions, keyword evaluator v2). All three models benefit from KG-RAG; deltas vary by model.

	Model	C1	C4g	Δ
Commercial	Claude Opus 4.6	21.8%	**69.0%**	+47.2 pp
Open	GPT-OSS 20B	20.5%	58.0%	+37.5 pp
Medical	MedGemma 27B	26.2%	57.8%	+31.5 pp

Evaluator bias. The keyword evaluator's abstention gate disproportionately penalizes C1, where the model correctly recognizes it lacks retrieval context and abstains. Physician review (Section 5.5) confirmed this bias: the auto-evaluator was too strict in 43% of cases (vs. too lenient in 10%), and the physician-rated C4g–C1 delta (+52 pp) exceeds the auto-evaluator delta (+47 pp), indicating the headline effect is conservative. An LLM-as-judge evaluation (Appendix V) provides a complementary perspective: the LLM judge achieves higher physician concordance (56.7% vs. 46.7%) and yields a C4g–C1 delta of +28.5 pp—lower than the keyword delta but still substantial. The asymmetry (C1 rises under LLM judge while C4g drops) is consistent with the keyword evaluator's conservative bias on abstentions.

5.2 Category×Condition Interaction

Table 3 and Figure 2 reveal a strong *category×condition interaction*: intent-aware KG-RAG improves all 9 categories over C1, with every delta exceeding +35 pp. Per-category results are diagnostic and exploratory; no multiplicity correction is applied. The preregistered primary endpoint is the hard longitudinal subset (change + current state + historical). Under the v2 evaluator with abstention detection, C1 drops sharply because the model without retrieval context frequently abstains ("insufficient information in the notes") rather than attempting an answer.

Where intent-aware routing excels. The largest gains occur on categories requiring cross-admission synthesis: change (+93 pp), sequence (+55 pp), historical (+54 pp), family history (+53 pp), conditional (+45 pp), and current state (+44 pp). On the hard longitudinal subset (change + current state + historical, $n = 130$), C4g outperforms C1 by +59.2 pp (95% BCa CI: [+49.2 pp, +67.7 pp]; patient-level: [+48.6 pp, +69.4 pp]). All nine category improvements exceed +35 pp (Appendix Figure 4). C6 fails catastrophically on historical (0%) and sequence (0%) despite Opus's 200K window, confirming that long context cannot substitute for structured retrieval on these categories.

5.3 Cross-Model Validation

All three models benefit substantially from KG-RAG (Table 4), with the magnitude consistent with an inverse relationship to parametric medical knowledge. Opus gains +47.2 pp, GPT-OSS gains +37.5 pp, and MedGemma—the only model fine-tuned on medical corpora—gains +31.5 pp. Notably, MedGemma starts with the highest C1 baseline (26.2%) yet achieves the lowest C4g accuracy (57.8%), while Opus starts at 21.8% but reaches 69.0% with retrieval.

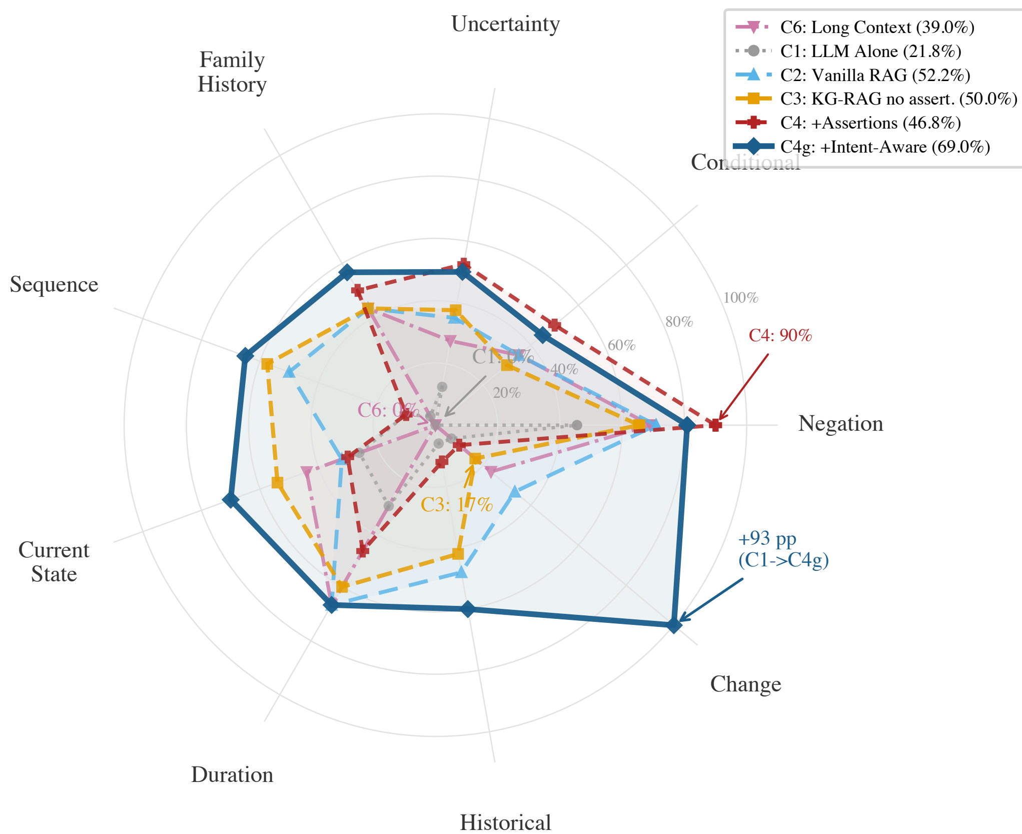

Figure 2: ClinicalBench per-category accuracy across five conditions (C6, C1, C2, C3, C4g; Claude Opus 4.6, evaluator v2). C6 (pink dash-dot) fails on historical and sequence (both 0%); C1 (gray dotted) falls below 50% on 7 of 9 categories; C2 (light blue) and C3 (orange dashed) score similarly overall (∼50–52%) but diverge per category; C4g (dark solid) adds epistemic annotations with intent routing. C4 (not shown; see Table 3) adds assertions without routing and scores 46.8%—below C3—demonstrating that intent routing, not assertions alone, drives the C3→C4g improvement.

Table 5: SliceBench overall results (144 questions, Claude Sonnet 4.5, judged by Opus 4.6). Sig: * denotes CI excluding zero.

	Condition	Score	95% CI	Δ vs B0
B0	LLM Alone	49.9%	[43.6%, 56.1%]	—
B1	Latest Note	73.4%	[67.8%, 78.7%]	+23.5 pp
B2	All Notes RAG	84.7%	[80.2%, 88.8%]	+34.8 pp *
B3	KG-RAG	86.9%	[82.8%, 90.8%]	+37.0 pp *
B4	Full System	87.6%	[83.7%, 91.2%]	+37.8 pp *

All three models improve on all 9 categories under C4g (per-category cross-model breakdowns in Appendix Table 9). Intent-aware KG-RAG thus functions as a *medical knowledge compensator*: the structured epistemic context partially substitutes for parametric medical knowledge, providing the largest benefit to general-purpose models that need it most.

5.4 SliceBench

The incremental KG layer (B2→B3) contributes +2.2 pp overall (CI: [-1.5 pp, +5.9 pp]), not reaching aggregate significance, but tier-stratified estimates show a complexity-dependent trend: +0.6 pp (Tier A), +1.0 pp (Tier B), +5.0 pp (Tier C, 15+ encounters), consistent with KG context being most valuable when document volume overwhelms chunk-level retrieval (Appendix L).

Table 6: Physician adjudication pilot ($n = 30$). Single reviewer, blinded during scoring; condition labels unblinded post-review for stratified analysis.

Metric	C1 ($n = 14$)	C4g ($n = 16$)
Physician-rated correct	29%	81%
Partially correct	21%	6%
Incorrect	50%	12%
Safe	50%	88%
Minor concern	50%	12%
Potentially harmful	0%	0%
Helpful	29%	81%
Neutral	29%	12%
Not useful	21%	0%
Misleading	21%	6%

5.5 Physician Adjudication

A board-certified physician (A.S.) reviewed a stratified pilot sample ($n = 30$ ClinicalBench items, all 9 categories) under blinded conditions (condition labels randomized as A/B; unblinding performed post-review for analysis). Each item was rated on four dimensions: gold-standard correctness, model-answer correctness, clinical safety, and clinical utility. Full rubric definitions and the scoring interface are described in Appendix O.

Physician-rated accuracy. After unblinding, KG-RAG answers (C4g, $n = 16$) were rated correct in 81% of items vs. 29% for LLM-alone (C1, $n = 14$; Table 6), a $+52$ pp physician-rated delta—larger than the $+47$ pp auto-evaluator delta and directionally consistent.

Safety and utility. C4g answers were rated safe in 88% of cases (vs. 50% for C1), helpful in 81% (vs. 29%), and misleading in 6% (vs. 21%). No answers in either condition were rated potentially harmful.

Auto-evaluator concordance. The keyword evaluator agreed with physician judgment in 47% of items. Disagreement was asymmetric: in 43% of cases the auto-evaluator was too strict (scored incorrect when the physician rated correct or partially correct), while in only 10% was it too lenient. This conservative bias is consistent with the abstention penalty noted in Section **??**: C1 answers that correctly decline to answer without context are penalized by the keyword matcher.

Gold-standard quality. Physician review found 40% of auto-generated gold standards required revision (30% incorrect, 10% needing revision), with errors concentrated in change detection (0/4 correct) and negation (0/2 correct) categories. Gold-standard errors are constant across conditions and therefore do not affect the relative C1–C4g comparison, but they indicate that auto-generated clinical benchmarks require physician validation—a finding with implications beyond this study.

Limitations. The pilot is limited by small per-condition samples ($n = 14/16$) and a single reviewer; a second independent reviewer is planned. Despite these constraints, the physician-rated delta ($+52$ pp) is directionally and magnitude-consistent with the automated delta ($+47$ pp) across all four dimensions measured.

6 Discussion and Limitations

Why long context underperforms. Brute-force long context (C6) scores 39.0%, outperforming C1 ($+17.2$ pp) because it at least provides clinical text for the model to draw on, but it still falls far short of C4g (69.0%) and fails completely on historical (0%) and sequence (0%). C6 concatenates all notes chronologically without retrieval or selection; more sophisticated long-context strategies (hierarchical summarization, temporal segmentation) might perform better, but C6's category-level failures—consistent with the "lost in the middle" phenomenon [37]—demonstrate that raw context dumping cannot substitute for structured retrieval. Historical questions require distinguishing resolved from active conditions across admissions; sequence questions require strict temporal ordering—both are implicit in raw text but explicit in the epistemic KG. C6 handles change detection modestly (23%) because the relevant information appears explicitly in the text, but KG-RAG achieves 100% (+77 pp).

Model-dependent KG benefit. Across the three models tested, KG-RAG benefit is consistent with an inverse relationship to parametric medical knowledge (Table 4). The model with the lowest C1 baseline (Opus, 21.8%) achieves the highest C4g accuracy (69.0%), while the medically fine-tuned model (MedGemma) starts highest (26.2%) but gains least under C4g. The cross-model pattern (Table 4) reinforces the knowledge-compensator interpretation: structured epistemic context partially substitutes for parametric knowledge, benefiting general-purpose models most. This has practical implications for RAG system design: the same retrieval pipeline may yield different gains depending on the downstream model, and single-model evaluation risks overstating or understating value.

When KG context matters. The complexity-dependent SliceBench effect (Tier C: $+5.0$ pp vs. Tier A: $+0.6$ pp) suggests structured epistemic context is most valuable when: (a) document volume overwhelms chunk-level retrieval, (b) questions require cross-encounter synthesis where the same concept appears with different assertion states, and (c) assertion status evolves across encounters.

The C4 ablation cleanly decomposes the C3→C4g gap (+19.0 pp) into two components: assertions alone (C3→C4: -3.2 pp, CI crossing zero) and intent routing (C4→C4g: $+22.2$ pp, $p < 10^{-10}$). This reveals a key architectural insight: assertion metadata is a necessary but insufficient condition—its value is unlocked only when paired with question-type-specific retrieval. Without routing, assertions increase evidence verbosity and degrade temporal categories (historical: -30 pp, sequence: -47.5 pp); with routing, the same metadata enables near-perfect change detection (C4 10% $\rightarrow$ C4g 100%). The finding validates the two-component design and suggests that future work on assertion-aware clinical NLP should couple enriched metadata with structured retrieval strategies.

Uncertainty remains weakest. Uncertainty improves from 12.5% to 50.0% (+37.5 pp), but absolute accuracy remains low. Possible and suspected conditions are linguistically subtle ("consider statin," "may have CHF") and require pragmatic inference beyond keyword matching.

Limitations. Sample size is modest: ClinicalBench covers 43 patients (400 questions) and SliceBench 6 patients (144 questions), both from MIMIC-IV [33]. The B2→B3 CI crosses zero ([-1.5 pp, +5.9 pp]); tier-level estimates ($n = 2$ patients per tier) are hypothesis-generating only. Model-dependent effects mean headline gains are specific to Opus; gains differ across models (Table 4). Physician review (Section 5.5) found the keyword evaluator agreed with physician judgment in only 47% of cases, predominantly erring on the conservative side (43% too strict vs. 10% too lenient); the physician-rated C4g–C1 delta (+52 pp) exceeds the auto-evaluator delta (+47 pp), suggesting the headline effect may be understated. An LLM-as-judge evaluation (Appendix V) achieves higher physician concordance (57% vs. 47%) and yields a C4g–C1 delta of +28.5 pp; the lower LLM-judge delta reflects the evaluator's capacity to credit partial answers and appropriate abstentions, particularly for C1. Gold-standard error rate was 40%, concentrated in change detection and negation; a single physician reviewed the pilot and a second reviewer is planned. This is an ablation study; direct comparisons with medical RAG systems [5, 6, 15] were not performed because these target factoid QA rather than patient-level longitudinal reasoning. C2 (vanilla TF-IDF document retrieval, 52.2%) serves as the closest proxy for these systems on our task, demonstrating that vanilla retrieval reaches roughly the same aggregate accuracy as graph-augmented retrieval without assertions (C3, 50.0%) but both fall ∼17–19 pp short of epistemic KG-RAG (C4g, 69.0%). Patient-level cluster bootstrap CIs (Appendix Table 16) are slightly wider than question-level CIs but all primary endpoints remain significant; the 43-patient sample nonetheless limits generalization. MedGemma run-to-run variance (±10 pp) arises from GPU non-determinism in 4-bit quantized inference. Over-reliance on KG-driven summaries risks anchoring bias if NLP extraction introduces systematic errors; single-site evaluation (US academic center) leaves generalization unvalidated (Appendix X).

Conclusion. The epistemic propagation gap is a real and measurable phenomenon. EpiKG addresses it through end-to-end epistemic preservation combined with intent-aware retrieval that matches graph traversal to question semantics. The ablation ladder decomposes the gain into three independent contributions: structured retrieval (+28 pp, C1→C3), assertion metadata (−3 pp without routing, C3→C4), and intent-aware routing (+22 pp, C4→C4g). The counterintuitive C4 result—that assertions alone degrade performance—reveals that the value of epistemic metadata is contingent on retrieval architecture, validating the two-component design. The benefit is both category-specific (every category improves, with the largest gains on longitudinal reasoning) and model-dependent (consistent across three models). Neither more context nor structured data alone suffices—C6 and C7 bracket the design space—confirming that the category×condition interaction, not the aggregate, is the right lens for clinical KG-RAG evaluation.

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A Notation Summary

Symbol	Meaning
$\alpha \in \mathcal{A}$	Assertion label (7 values, Eq. 1)
$\boldsymbol{\tau}_v$	Valid time (event date, valid from/to)
$\boldsymbol{\tau}_t$	Transaction time (recorded at, doc date, created at)
τ_a	NLP-asserted temporality (Current / Past / Future)
$r \in \mathcal{R}$	Temporal relation (9 values, Table 19)
c	Confidence score $\in [0, 1]$
ξ	Experiencer (Patient / Family)
$e(m)$	Epistemic state tuple (c, α, ξ, τ)
q, π	Clinical question, patient
ι	Question intent (Change / Current_State / Historical / Default)
$f_{np}(c)$	Fraction of non-present assertions for concept c

Table 7: Key notation used throughout the paper.

B Formal Epistemic Preservation

We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the principle that aboutness—the relationship between information artifacts and the entities they represent—must be preserved across transformations [29].

Definition 2 (Epistemic State)**.** *For a clinical mention m, the* epistemic state *is the tuple $e(m) = (c, \alpha, \xi, \tau)$, where c is the OMOP concept identifier, $\alpha \in \mathcal{A}$ is the 7-value assertion label (Eq. 1), $\xi \in \{\text{PATIENT}, \text{FAMILY}\}$ is the experiencer, and $\tau \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ is the temporality. A pipeline P* epistemically preserves *mention m if $P(e(m)) = e(m)$; that is, the epistemic state at the output of every pipeline stage is identical to the state at extraction.*

Proposition 3 (Assertion Entropy Loss)**.** *For concept c in patient π's record, let A_c^π denote the assertion distribution with empirical frequencies $p(\alpha_i) = n_i / \sum_j n_j$ over the $|\mathcal{A}|$ assertion classes. The assertion entropy is*

$$H(A_c^\pi) = -\sum_i p(\alpha_i) \log p(\alpha_i) \ . \tag{3}$$

An assertion-blind pipeline collapses all mentions to $\alpha = \text{PRESENT}$, yielding a degenerate distribution with $H = 0$. The information loss is $\Delta H = H(A_c^\pi) \geq 0$ [30], with $\Delta H > 0$ strictly whenever any mention carries a non-present assertion.

Proof. Under the assertion-blind mapping $\phi : \alpha_i \mapsto \text{PRESENT}$ for all i, the output distribution assigns probability 1 to PRESENT and 0 to all other classes, so $H(\phi(A_c^\pi)) = 0$. By non-negativity of Shannon entropy, $\Delta H = H(A_c^\pi) - 0 = H(A_c^\pi) \geq 0$, with equality iff all mentions already carry $\alpha = \text{PRESENT}$. □

Corollary 4 (Faithfulness Bound)**.** *Let $f_{np}(c)$ denote the fraction of mentions of concept c carrying a non-present assertion ($\alpha \neq \text{PRESENT}$). Without assertion labels, the maximum assertion-faithful accuracy for any downstream task conditioned on c is bounded by $1 - f_{np}(c)$. In clinical records where negated and uncertain mentions are prevalent—e.g., "no pneumonia," "possible CHF"—this bound can be substantially below 1.*

Proof. Without assertion labels, any predictor must assign a single assertion class to all mentions of c. Choosing PRESENT (the majority class in clinical text) yields accuracy $1 - f_{np}(c)$; the $f_{np}(c)$ non-present mentions are necessarily misclassified. □

C Gap Analysis

Partial marks indicate limited coverage: GFM-RAG, MedRAG, and KARE build population-level or hierarchical KGs (not per-patient longitudinal graphs); KARE uses KG-augmented retrieval but

Table 8: Capability comparison across representative systems. ✓ = supported, ○ = partial, × = not supported.

System	Patient KG	OMOP mapping	Assertion (7-class)	Tri-temporal	Assert.-aware RAG	Experiencer	Allen's algebra	Multi-hop (≥3)
DoctorRAG [15]	×	×	×	×	×	×	×	×
GFM-RAG [5]	○	×	×	×	×	×	×	✓
MedRAG [16]	○	×	×	×	×	×	×	○
KARE [6]	○	×	×	×	○	×	×	✓
Multi-LLM KG [20]	×	✓	○	×	×	×	×	×
MedTKG [10]	✓	×	×	○	×	×	×	✓
Graphiti [11]	×	×	×	○	×	×	×	✓
MediGRAF [18]	✓	×	×	×	×	×	×	✓
EpiKG (ours)	✓	✓	✓	✓	✓	✓	✓	○

without assertion awareness; Multi-LLM KG models uncertainty via entropy but not the full assertion taxonomy; MedTKG and Graphiti use temporal annotations but not tri-temporal models. MediGRAF is the closest patient-level competitor, constructing per-patient medical graphs from clinical notes, but it does not preserve assertion status or temporal relations as edge attributes. EpiKG's partial mark on multi-hop traversal ($\geq$ 3 hops) reflects a deliberate architectural trade-off: PostgreSQL CTE-based traversal provides ACID compliance but degrades beyond 2 hops (Appendix K).

D Safety Score

The safety score $S = 1 - \frac{1}{N}\sum_{i=1}^{N} w_i \cdot \mathbb{1}[\hat{y}_i \neq y_i]$ applies $w_i = 2.0$ for false-positive assertion errors (reporting a negated or absent condition as present) and $w_i = 1.0$ otherwise, capturing asymmetric clinical risk where acting on a falsely affirmed condition is more dangerous than missing a present one. We treat $w = 2.0$ as a reasonable default; sensitivity to this choice is a direction for future work.

E Cross-Model Per-Category Results

Table 9: ClinicalBench per-category accuracy (%) by model under C1 and C4g (intent-aware KG-RAG, evaluator v2). All three models improve on all 9 categories.

		Claude Opus 4.6			MedGemma 27B			GPT-OSS 20B		
Category	n	C1	C4g	Δ	C1	C4g	Δ	C1	C4g	Δ
Negation	110	45.5	80.9	↑35.5 pp	57.3	76.4	↑19.1 pp	46.4	80.0	↑33.6 pp
Conditional	20	0.0	45.0	↑45.0 pp	15.0	35.0	↑20.0 pp	0.0	15.0	↑15.0 pp
Uncertainty	40	12.5	50.0	↑37.5 pp	7.5	32.5	↑25.0 pp	5.0	30.0	↑25.0 pp
Family hist.	30	3.3	56.7	↑53.3 pp	3.3	40.0	↑36.7 pp	3.3	40.0	↑36.7 pp
Sequence	40	10.0	65.0	↑55.0 pp	7.5	47.5	↑40.0 pp	12.5	42.5	↑30.0 pp
Current st.	50	26.0	70.0	↑44.0 pp	32.0	58.0	↑26.0 pp	36.0	56.0	↑20.0 pp
Duration	30	30.0	66.7	↑36.7 pp	53.3	70.0	↑16.7 pp	16.7	70.0	↑53.3 pp
Historical	50	6.0	60.0	↑54.0 pp	0.0	46.0	↑46.0 pp	0.0	60.0	↑60.0 pp
Change	30	6.7	100.0	↑93.3 pp	0.0	76.7	↑76.7 pp	0.0	70.0	↑70.0 pp
Overall	**400**	**21.8**	**69.0**	**↑47.2** pp	**26.2**	**57.8**	**↑31.5** pp	**20.5**	**58.0**	**↑37.5** pp

F MedGemma Full Per-Category Results

Category	n	**C7**	**C1**	**C4g**
Negation	110	13.6	57.3	76.4
Conditional	20	0.0	15.0	35.0
Uncertainty	40	0.0	7.5	32.5
Family History	30	0.0	3.3	40.0
Sequence	40	0.0	7.5	47.5
Current State	50	0.0	32.0	58.0
Duration	30	0.0	53.3	70.0
Historical	50	0.0	0.0	46.0
Change	30	0.0	0.0	76.7
Overall	**400**	**3.8**	**26.2**	**57.8**

Table 10: MedGemma 27B ClinicalBench per-category accuracy (%, keyword evaluator v2) for three conditions. MedGemma gains +32 pp overall under C4g, with all 9 categories improving. Strongest improvements on change (+77 pp) and historical (+46 pp).

G Experiencer Attribute Propagation Ablation

The `experiencer` attribute distinguishes patient conditions from family member conditions; without it, the graph conflates the two, causing family history misattribution.

Table 11: Impact of experiencer attribute propagation on Opus C4 (400 questions, prior evaluator run). Categories with no change omitted.

Category	Pre-fix	Post-fix	Δ
Family history	63.3%	**73.3%**	+10.0 pp
Uncertainty	50.0%	**55.0%**	+5.0 pp
Historical	34.0%	38.0%	+4.0 pp
Current state	46.0%	48.0%	+2.0 pp
Overall C4	**68.2%**	**70.2%**	+2.0 pp

The fix improved family history by +10.0 pp with zero regressions on guard categories (negation, conditional, duration, sequence all unchanged), confirming that the experiencer attribute is load-bearing for assertion-sensitive reasoning.

H SliceBench Figures

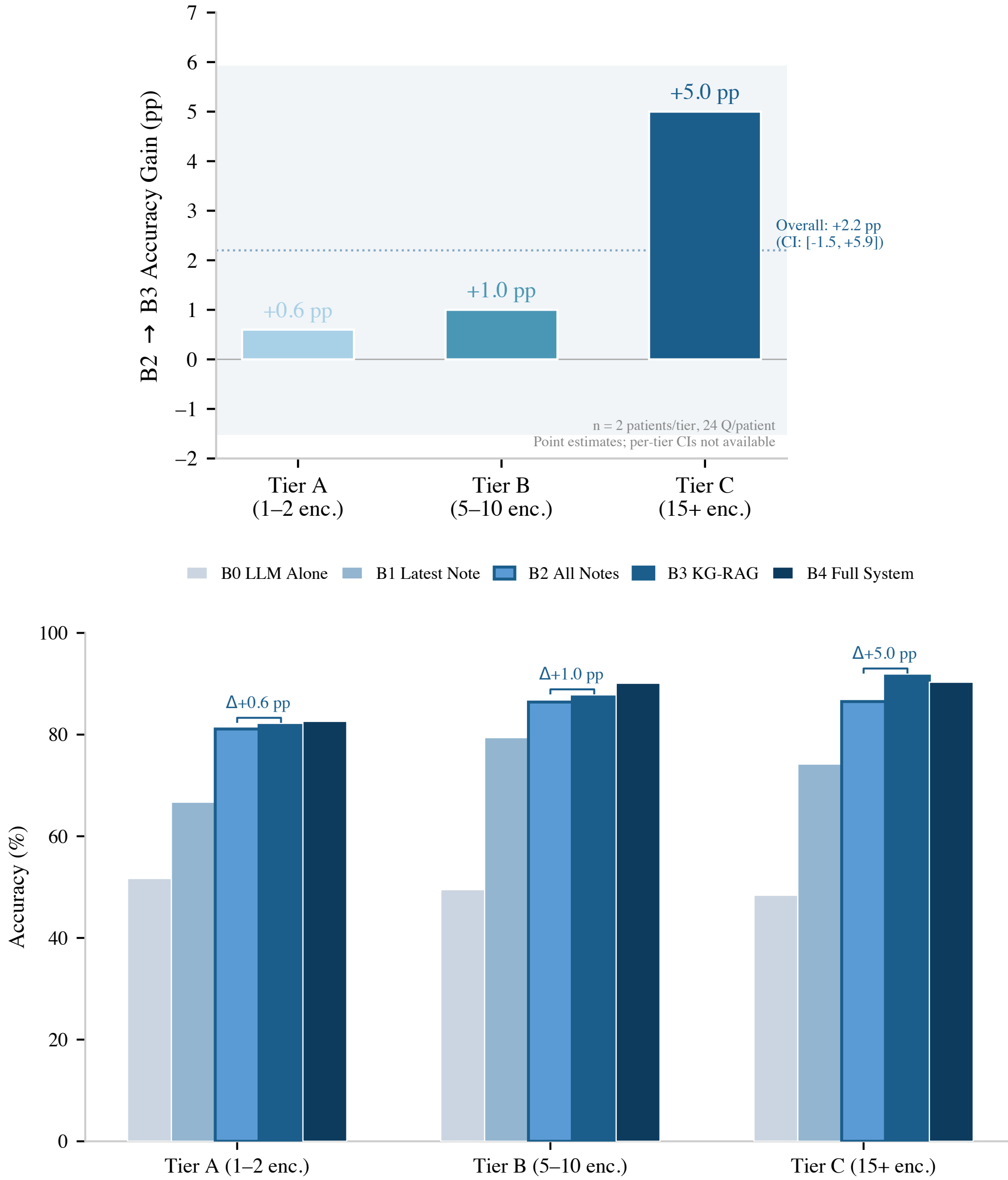

Figure 3: SliceBench results ($n = 2$ patients per tier, 24 questions each; point estimates without per-tier CIs). **Top:** The KG layer's incremental benefit (B2→B3) shows a directional, complexity-dependent trend: $+0.6$ pp (Tier A) to $+5.0$ pp (Tier C). **Bottom:** Full condition progression by tier. The overall B2→B3 delta is $+2.2$ pp (CI: [-1.5 pp, +5.9 pp]).

I Per-Category Delta Chart

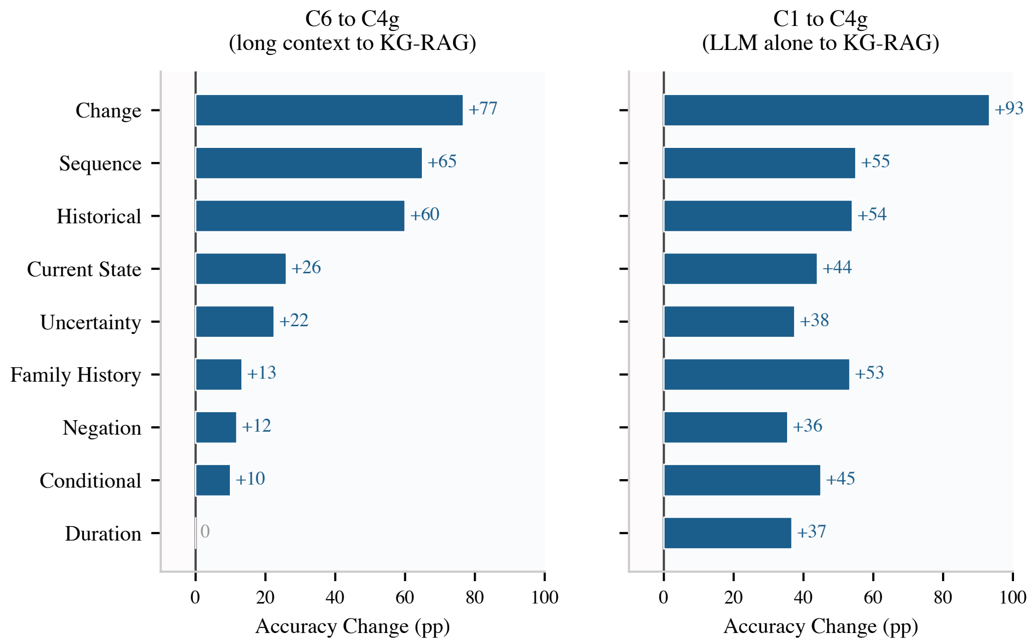

Figure 4: Per-category accuracy change (Claude Opus 4.6, evaluator v2) for two comparisons. **Left:** C6→C4g (structured vs. unstructured retrieval): KG-RAG outperforms long context on all 9 categories, with the largest gains on sequence (+65 pp) and historical (+60 pp). **Right:** C1→C4g (LLM alone to KG-RAG): all 9 categories improve—change (+93 pp), sequence (+55 pp), historical (+54 pp).

J SliceBench Conditions

Table 12: SliceBench conditions. B0–B4 form a monotone context progression.

ID	Condition	Context Provided to LLM
B0	LLM Alone	Question only (parametric knowledge)
B1	Latest Note	Most recent clinical note
B2	All Notes RAG	All notes, retrieved by relevance
B3	KG-RAG	Knowledge graph context + all notes
B4	Full System	KG-RAG + guidelines + calculators

K System Implementation Details

LLM baseline. On MedQA-USMLE (965 questions), Claude Opus 4.5 achieves 81.6% accuracy (5.1 pp below GPT-4 at 86.7% and Med-PaLM 2 at 86.5% [32]), establishing that the Claude model family—Sonnet 4.5 (SliceBench answerer) and Opus 4.6 (ClinicalBench primary, SliceBench judge)—provides a reasonable LLM baseline (Table 13).

Model	**Accuracy**	**Step 1**	**N**
EpiKG (LLM alone)	81.6%	79.9%	965
GPT-4 (2023)	86.7%	—	1,273
Med-PaLM 2 (2023)	86.5%	—	1,273
Claude 3 Opus (2024)	78.0%	—	1,273

Table 13: MedQA-USMLE results. Our LLM-alone baseline is competitive.

KG scale and traversal latency. The deployed system processes 145 documents across 85 patients, materializing 3,100 KG nodes and 8,803 edges. Graph traversal latency is sub-millisecond at this scale: 0.57 ms (1-hop), 0.75 ms (2-hop). The full system integrates 201 clinical calculators, 1,202 guideline sections, 20M+ OMOP vocabulary relationships, 122 assertion trigger patterns, and 24 edge types across 13 node types.

Multi-hop traversal. On the DR.KNOWS diagnostic reasoning benchmark [38], which measures KG traversal accuracy using PostgreSQL-backed clinical data, EpiKG achieves 0.420 overall (50% at 1-hop, 25% at 2-hop, 0% at 3-hop). Multi-hop degradation reflects a deliberate architectural trade-off: PostgreSQL CTE-based traversal provides ACID compliance but scales poorly beyond 2 hops.

L SliceBench Tier-Stratified Results

Table 14: SliceBench results stratified by patient complexity tier. B2→B3 Δ shows the incremental KG contribution.

		Score (%)					
Tier	Encounters	B0	B1	B2	B3	B4	B2→B3 Δ
A	1–2 notes	51.7	66.7	81.1	81.8	82.6	+0.6 pp
B	5–10 notes	49.5	79.4	86.4	87.4	90.1	+1.0 pp
C	15+ notes	48.4	74.2	86.5	91.5	90.3	+5.0 pp

M ClinicalBench Full Per-Category Results (Opus)

Category	n	**C7**	**C6**	**C1**	**C2**	**C3**	**C4**	**C4g**
Negation	110	13.6	69.1	45.5	70.9	65.5	90.0	80.9
Conditional	20	0.0	35.0	0.0	35.0	30.0	50.0	45.0
Uncertainty	40	0.0	27.5	12.5	35.0	37.5	52.5	50.0
Family History	30	0.0	43.3	3.3	43.3	43.3	50.0	56.7
Sequence	40	0.0	0.0	10.0	60.0	57.5	10.0	65.0
Current State	50	0.0	44.0	26.0	32.0	54.0	30.0	70.0
Duration	30	0.0	66.7	30.0	66.7	60.0	46.7	66.7
Historical	50	0.0	0.0	6.0	48.0	42.0	12.0	60.0
Change	30	0.0	23.3	6.7	43.3	16.7	10.0	100.0
Overall	**400**	**3.8**	**39.0**	**21.8**	**52.2**	**50.0**	**46.8**	**69.0**

Table 15: Complete ClinicalBench per-category accuracy (%, Claude Opus 4.6, keyword evaluator v2) for all conditions. C4 adds assertion metadata to C3 without intent routing; it helps assertion-sensitive categories (negation: +24.5 pp) but hurts temporal categories (historical: −30 pp, sequence: −47.5 pp). Only with intent routing (C4g) does the full system reach 69%.

N Physician Validation Pilot

A board-certified physician (A.S.) reviewed a stratified sample of $n = 30$ ClinicalBench items (14 C1, 16 C4g) spanning all 9 categories under blinded conditions. Each item was rated on: (1) gold-standard correctness (Yes / Partially / Needs Revision / No), (2) model-answer correctness (Correct / Partially Correct / Incorrect), (3) clinical safety (Safe / Minor Concern / Potentially Harmful), and (4) clinical utility (Helpful / Neutral / Not Useful / Misleading). Reviews were conducted using a custom scoring interface with access to the full de-identified MIMIC-IV discharge summaries, filtered to the admission(s) relevant to each question.

Gold-standard error analysis. Of the 30 gold standards reviewed, 18 (60%) were rated correct, 3 (10%) needed revision, and 9 (30%) were incorrect. Errors were concentrated in change detection (0/4 gold standards correct—NLP misread discharge medication lists) and negation (0/2 correct—NLP missed physical exam findings). These errors are constant across conditions and do not affect the relative C1–C4g comparison.

Auto-evaluator concordance. The keyword evaluator agreed with physician judgment in 47% of cases. In 43% of disagreements the evaluator was too strict (marked incorrect when the physician rated correct or partially correct); in 10% it was too lenient. This conservative bias largely reflects the abstention penalty: C1 answers that appropriately decline to answer without patient context are penalized by keyword matching.

O Physician Validation Protocol

Design. A board-certified physician independently scored ClinicalBench answers under two conditions (C1 and C4g) using a multi-dimensional rubric (correctness, safety, utility). The reviewer was blinded to condition assignment (conditions randomized as A/B). The pilot sample comprises 30 items stratified across all 9 categories and both conditions, selected to maximize coverage within a feasible review budget ($\sim$8 hours physician time).

Endpoints. (1) Human–keyword evaluator concordance rate (fraction of automated scores confirmed by physician). (2) Physician-rated C4g vs. C1 accuracy, safety, and utility. (3) Gold-standard error rate.

Disclosure. The reviewing physician (A.S.) is a co-author. Single-reviewer design limits inter-rater reliability assessment; a second reviewer is planned for the extended validation.

Status. This protocol was designed before examining primary results. Pilot results ($n = 30$) are reported in Section 5.5.

P Reproducibility Details

Models. ClinicalBench primary ablation: Claude Opus 4.6 (claude-opus-4-20250514, keyword evaluator v2). Cross-model: MedGemma 27B (gemma3:27b, 4-bit GGUF) and GPT-OSS 20B (4-bit GGUF) via Ollama; same evaluator. SliceBench: Claude Sonnet 4.5 (claude-sonnet-4-5-20250929) answers, Opus 4.6 judges. Temperature 0 throughout; 4-bit quantization introduces GPU non-determinism ($\pm$10 pp run-to-run for MedGemma), controlled via within-run paired comparisons.

Evaluation. The deterministic keyword evaluator performs exact word-boundary matching (regex `\bKEYWORD\b`) against gold-standard assertion and temporal keywords. Per-category keyword sets: negation ("no," "denies," "absent," "negative"), uncertainty ("possible," "suspected," "may," "likely," "consider"), temporal ("before," "after," "during," "changed," "new"), plus category-specific terms. Evidence preambles (echoed graph context) are stripped before matching.

Retrieval. C2's vanilla RAG uses TF-IDF over chunked clinical notes (512-token chunks, 64-token overlap), retrieving the top-5 chunks by cosine similarity. This is the closest analog to existing medical RAG systems (DoctorRAG, MedRAG) on patient-level longitudinal data. ClinicalBench conditions map to SliceBench: C1≈B0, C2≈B2, C4g≈B3, C5≈B4.

Bootstrap. $n = 2{,}000$ resamples, seed 42, BCa method [35]. We report both question-level and patient-level (cluster) bootstrap CIs; the latter resample the 43 patients with replacement, including all their questions per draw, to account for within-patient correlation. Patient-level CIs are slightly wider but all primary endpoints remain significant (Table 16).

McNemar's test. We supplement bootstrap CIs with McNemar's test for paired nominal data, comparing discordant pairs (C1 wrong/C4g right vs. C1 right/C4g wrong). All three pairwise comparisons are significant: C1 vs. C4g ($\chi^2 = 157.1$, $p < 10^{-6}$; discordant: 18 vs. 207), C1 vs. C3 ($\chi^2 = 73.4$, $p < 10^{-6}$; 29 vs. 142), and C3 vs. C4g ($\chi^2 = 49.3$, $p < 10^{-11}$; 19 vs. 95).

Data. De-identified MIMIC-IV clinical notes accessed under a PhysioNet Credentialed Health Data Use Agreement (v1.5.0).

Compute. ClinicalBench Opus C1/C3/C4g: $\sim$22min each (API); Opus C6: $\sim$3.5h (API); C7: $<$ 1min (no LLM); MedGemma C1–C4g: $\sim$3.6h (single GPU); MedGemma C6: $\sim$1.5h; SliceBench: $\sim$2h (API).

Table 16: Statistical summary: question-level and patient-level (cluster) BCa bootstrap 95% CIs for primary endpoints. Both levels exclude zero for all comparisons.

Comparison	Δ	Question CI	Patient CI	McNemar p
C4g−C1 (overall)	+47.2 pp	[+40.9 pp, +52.8 pp]	[+40.2 pp, +52.5 pp]	$< 10^{-6}$
C3−C1 (retrieval)	+28.2 pp	[+22.3 pp, +34.0 pp]	[+23.1 pp, +33.8 pp]	$< 10^{-6}$
C4−C3 (assertions)	−3.2 pp	[-10.0 pp, +3.2 pp]	[-10.8 pp, +3.4 pp]	0.37
C4g−C4 (routing)	+22.2 pp	[+16.0 pp, +28.0 pp]	[+14.8 pp, +29.3 pp]	$< 10^{-10}$
C4g−C3 (both)	+19.0 pp	[+13.8 pp, +24.0 pp]	[+14.7 pp, +23.5 pp]	$< 10^{-11}$
Hard longitudinal	+59.2 pp	[+49.2 pp, +67.7 pp]	[+48.6 pp, +69.4 pp]	—

Evaluator evolution. The keyword evaluator underwent three versions: v0 (substring matching), v1 (word-boundary matching + evidence preamble stripping), and v2 (+ abstention detection gate + domain-specific keyword requirements for sequence and change). The v1 evaluator awarded false positives when models responded with "insufficient information in the notes" because negation and temporal keywords in the refusal text matched gold-standard patterns. The v2 evaluator adds an abstention detection layer: answers matching abstention patterns (e.g., "cannot determine," "not mentioned," "insufficient evidence") are scored as incorrect unless they contain clinical claim patterns (e.g., "patient does not," "denies"). Additionally, v2 requires sequence answers to contain ordering keywords ("first," "then," "before") and change answers to contain change keywords ("added," "removed," "discontinued"), preventing term-overlap-only false positives. The duration category's minimum-0.5 score floor for matching duration keywords was removed. The v1→v2 transition reduced C1 accuracy (from ∼50% to 21.8% for Opus) by correctly classifying abstention responses as incorrect; C4g accuracy decreased modestly (from ∼76% to 69.0%) because the model abstains less frequently when retrieval context is provided. All main-text numbers use v2.

Reproducibility package. A standalone reproducibility package is included as supplementary material (`epikg-benchmark/`). It contains all 400 ClinicalBench questions with gold answers, raw model predictions for all condition×model combinations (Opus C1/C3/C4g/C6, MedGemma C1/C4g, GPT-OSS C1/C4g), and the deterministic keyword evaluator v2 with abstention detection. Running the evaluator requires only Python 3.10+ with no external dependencies. All accuracy numbers in Tables 2, 3, and 4 are independently reproducible from this package. MIMIC-IV patient identifiers are included so that PhysioNet-credentialed reviewers can trace predictions back to source clinical notes.

Results provenance. Table 17 maps each reported result to its source checkpoint file. MedGemma C4g has 399 predictions (1 timeout); the missing question is scored as incorrect ($n = 400$).

Table 17: Results provenance: checkpoint files for each condition×model combination. All scored with keyword evaluator v2.

Condition	Model	Checkpoint file	n
C1	Opus 4.6	`opus/C1_llm_alone.json`	400
C2	Opus 4.6	`opus/C2_vanilla_rag.json`	400
C3	Opus 4.6	`opus/C3_kg_rag.json`	400
C4	Opus 4.6	`opus/C4_epistemic_kg_rag.json`	400
C4g	Opus 4.6	`opus/C4g_intent_aware.json`	400
C6	Opus 4.6	`opus/C6_long_context.json`	400
C7	—	`opus/C7_deterministic.json`	400
C1	MedGemma 27B	`medgemma/C1_llm_alone.json`	400
C4g	MedGemma 27B	`medgemma/C4g_intent_aware.json`	399*
C1	GPT-OSS 20B	`gptoss/C1_llm_alone.json`	400
C4g	GPT-OSS 20B	`gptoss/C4g_intent_aware.json`	400

*MedGemma C4g: 1 timeout; missing answer scored as incorrect for $n = 400$ denominator.

Q Assertion Category Definitions

Assertion	Definition	Example
PRESENT	Condition currently affirmed	“has diabetes”
ABSENT	Condition explicitly negated	“denies chest pain”
POSSIBLE	Condition suspected, not confirmed	“possible pneumonia”
CONDITIONAL	Contingent on specific circumstances	“if febrile, start antibiotics”
HYPOTHETICAL	Discussed as a scenario	“would need dialysis if…”
FAMILY_HISTORY	Attributed to a family member	“mother had breast cancer”
HISTORICAL	Previously true, not necessarily current	“former smoker”

Table 18: The 7-value assertion taxonomy, extending i2b2 by separating HISTORICAL from FAMILY_HISTORY.

R Temporal Relation Mapping

The nine temporal relations $\mathcal{R}$ used on KG edges are derived from Allen’s 13 canonical interval relations [12] by merging symmetric pairs.

$\mathcal{R}$ value	Allen source(s)	Meaning
BEFORE	Before, Meets	A ends before B starts
AFTER	After, Met-by	A starts after B ends
DURING	During	A entirely within B
CONTAINS	Contains	A entirely contains B
OVERLAPS	Overlaps, Overlapped-by	A and B partially overlap
STARTS	Starts, Started-by	A and B share start time
FINISHES	Finishes, Finished-by	A and B share end time
CONCURRENT	Equals	A and B have same interval
UNKNOWN	—	Relation undetermined

Table 19: Mapping from Allen’s 13 interval relations to the 9 temporal relation values stored on KG edges.

S Intent-Aware Routing Algorithm

The C4g classifier is rule-based, mapping questions to one of four intents using keyword patterns (e.g., “changed”, “new since” trigger CHANGE; “currently”, “active problem” trigger CURRENT_STATE). Algorithm 1 details the full retrieval procedure.

Algorithm 1: Intent-Aware Retrieval (C4g)

Input: Question q, patient π
Output: Structured evidence E

$\mathcal{C} \leftarrow \textsc{ExtractConcepts}(q)$; `// NLP + OMOP enrichment`
$\iota \leftarrow \textsc{ClassifyIntent}(q)$; `//` $\in \{\textsc{Change}, \textsc{CurrSt}, \textsc{Hist}, \textsc{Default}\}$
if $\iota = \textsc{Change}$ **then**
  Partition edges by admission: $\mathcal{E}_k \leftarrow \{e \mid \texttt{hadm\_id}(e) = k\}$;
  foreach *admission pair* (k, k') *with* $k < k'$ **do**
    $\mathcal{A} \leftarrow \mathcal{C}_{k'} \setminus \mathcal{C}_k$; $\mathcal{R} \leftarrow \mathcal{C}_k \setminus \mathcal{C}_{k'}$; $\mathcal{S} \leftarrow \mathcal{C}_k \cap \mathcal{C}_{k'}$;
  $E_g \leftarrow \textsc{FormatChange}(\mathcal{A}, \mathcal{R}, \mathcal{S})$;
else if $\iota = \textsc{CurrSt}$ **then**
  $E_g \leftarrow \textsc{FilterEdges}(\pi, \mathcal{C},\ \tau_a = \textsc{Current} \vee \text{open validity})$;
  Deduplicate by concept; emit “NOT FOUND” for missing $c \in \mathcal{C}$;
else if $\iota = \textsc{Hist}$ **then**
  $E_g \leftarrow \textsc{FilterEdges}(\pi, \mathcal{C},\ \tau_a = \textsc{Past})$;
  Augment: concepts in earlier admissions but absent from latest $\rightarrow$ “resolved”;
else
  $E_g \leftarrow \textsc{BidirectionalBFS}(\pi, \mathcal{C},\ \text{hops} = 2\text{–}3,\ c_{\min} = 0.3)$;
$E_d \leftarrow \textsc{RetrieveDocuments}(\pi, \mathcal{C})$;
return $\textsc{Compose}(E_g, E_d,\ \text{template}(\iota))$;

T Bookend and Diagnostic Conditions

C5 (Full System). C5 extends C4 with guideline retrieval (1,202 sections) and clinical calculators (201, e.g., CHA_2DS_2-VASc, MELD). In a prior evaluator run, C5 achieved 68.2% overall, below the prior-run C1 and C4, likely because non-relevant components dilute the context window.

C6 (Long Context). All patient documents are concatenated chronologically and presented to the LLM. For Opus (200K token window), all notes fit; for MedGemma (8K window), later documents are truncated. C6 achieves 39.0% (Opus) overall—better than C1 (21.8%) but far below C4g (69.0%). Opus scores 0% on historical and sequence under C6, confirming that brute-force long context cannot substitute for structured retrieval on categories requiring cross-encounter synthesis.

C7 (Deterministic KG). KG edges matching query concepts are returned directly without LLM reasoning. C7 achieves 3.8% overall (only negation at 13.6% via stored assertion metadata), confirming that structured data without an LLM reasoner cannot answer clinical questions.

U Illustrative Retrieval Example

To illustrate why intent-aware routing matters, consider a change question: *"What medications changed between this patient's first and second admissions?"*

C1 (LLM alone). The model sees only the current note, which mentions metoprolol, atorvastatin, and "discontinued lisinopril." Without prior admission records, it cannot determine what was *added* vs. *continued*, and may hallucinate prior medication lists.

C4g (intent-aware KG-RAG). The intent classifier routes to CHANGE retrieval, which partitions KG edges by `hadm_id` and computes set differences:

- **Added** (admission 2 only): atorvastatin 40mg [PRESENT]
- **Removed** (admission 1 only): lisinopril 10mg [PRESENT→ABSENT]
- **Continued**: metoprolol 25mg [PRESENT, both admissions]

The assertion labels (PRESENT, ABSENT) disambiguate: "discontinued lisinopril" is not a current medication but a historical one whose status changed—exactly the distinction the epistemic schema preserves.

V LLM-as-Judge Concordance

To complement the deterministic keyword evaluator, we rescored all 800 predictions (400 questions $\times$ 2 conditions: C1 and C4g) using Claude Opus 4.6 as an LLM judge. The judge prompt presents the question, reference answer, and system answer, and requests a score of 1 (correct), 0.5 (partially correct), or 0 (incorrect) with a one-sentence justification. Table 20 summarizes the concordance.

Table 20: Evaluator concordance comparison. LLM judge scores ≥ 0.5 treated as correct for binary comparison. Physician concordance computed on $n = 30$ overlapping items.

Metric	Keyword v2	LLM Judge
vs. Keyword evaluator ($n = 800$)		
Agreement	—	78.9%
Cohen's κ	—	0.572
vs. Physician ($n = 30$)		
Agreement	46.7%	56.7%
Too strict	43.3%	36.7%
Too lenient	10.0%	6.7%
Condition-level accuracy		
C1 accuracy	21.8%	28.5%
C4g accuracy	69.0%	57.0%
C4g−C1 Δ	+47.2 pp	+28.5 pp

Key findings. The LLM judge achieves higher physician concordance (56.7% vs. 46.7%) and is less prone to false strictness (36.7% vs. 43.3%), confirming that the keyword evaluator's conservative bias inflates the measured delta. Under the LLM judge, C1 rises from 21.8% to 28.5% (the judge credits clinically reasonable hedging that keyword matching penalizes), while C4g drops from 69.0% to 57.0% (the judge penalizes partial answers that pass keyword matching via term overlap). The resulting C4g−C1 delta under LLM judge ($+28.5$ pp) is lower than the keyword delta ($+47.2$ pp) but remains substantial and directionally consistent. The per-condition asymmetry—C1 gains, C4g loses—is consistent with the physician finding that the keyword evaluator disproportionately penalizes C1 abstentions.

Per-category analysis. The LLM judge is particularly stricter on change (C4g: 62% mean score vs. 100% keyword) because it penalizes partial medication lists that contain correct change keywords but miss specific drugs. Conversely, the judge is more generous on historical (C4g: 62% vs. 60% keyword) and negation (C4g: 79% vs. 81% keyword), where its semantic understanding better captures correct answers that use varied phrasing.

Limitations. Using the same model family (Opus) for both answering and judging introduces potential self-preference bias. The judge also shows moderate agreement with the keyword evaluator ($\kappa = 0.572$), indicating they capture partially overlapping but distinct aspects of answer quality.

W Assertion Classifier Evaluation

The rule-based assertion classifier uses 122 calibrated trigger patterns extending the NegEx [24] and ConText [25] frameworks to a 7-class taxonomy (Table 18). Table 21 summarizes the pattern inventory and confidence ranges by category.

Table 21: Assertion trigger pattern inventory. Confidence ranges reflect per-trigger calibration.

Category	Patterns	Confidence	Example triggers
Absent	31	0.85–0.98	"no evidence of," "denies," "ruled out"
Uncertain	32	0.35–0.75	"possible," "likely," "consistent with"
Present	27	0.85–0.98	"confirmed," "diagnosed with," "positive for"
Hypothetical	11	0.25–0.35	"risk of," "screening for," "prophylaxis"
Family history	8	0.85–0.95	"family history of," "maternal history"
Conditional	7	0.20–0.40	"if," "contingent on," "depending on"
Historical	6	0.75–0.90	"history of," "former," "remote history"
Total	**122**	0.20–0.98	

Corpus statistics. Across the 43 ClinicalBench patients, the knowledge graph contains 3,943 edges linked to clinical facts. Of these, 618 (15.7%) carry non-present assertions: 442 absent (11.2%), 94 possible (2.4%), 71 historical (1.8%), 8 conditional (0.2%), and 3 hypothetical (0.1%). At the mention level, 1,428 of 12,379 mentions (11.5%) are non-present. This non-present fraction $f_{np} = 0.157$ provides the empirical bound from Corollary 4: an assertion-blind pipeline cannot exceed $1 - f_{np} = 84.3\%$ assertion-faithful accuracy on concepts where negated or uncertain mentions are present.

Literature-grounded accuracy. No annotated assertion corpus was available for intrinsic evaluation; we instead bound expected accuracy using published benchmarks on similar rule-based classifiers. NegEx achieves 94.5% precision / 77.8% recall for negation [24]; ConText reports 93.8% precision for negation scope [25]; NegBio reaches 96.3% precision / 85.7% recall [28]. Our classifier extends these with calibrated confidence, pseudo-negation handling, and scope-aware bidirectional matching. Expected accuracy is $\sim$85–95% for negation (the most studied and our largest non-present class at 71.5% of non-present edges) and $\sim$75–85% for uncertainty and historical (less studied, fewer trigger patterns).

Functional evaluation. Rather than relying solely on intrinsic metrics, the C4 ablation provides a functional evaluation of assertion quality. C4 adds assertion metadata to C3's graph without intent routing: the classifier's output is directly consumed by the retrieval pipeline. On assertion-sensitive categories (negation, conditional, uncertainty, family history), C4 outperforms C3 by an average of $+16.5$ pp, confirming that the classifier produces usable assertions for these categories. On temporal categories (historical, sequence, change, current state, duration), C4 underperforms C3 by an average of -24 pp—not because assertions are incorrect, but because uniform BFS traversal cannot exploit them effectively. The C4→C4g routing correction recovers these losses and amplifies the gains, achieving $+22.2$ pp overall ($p < 10^{-10}$).

X Ethics Statement

This work uses de-identified clinical data from MIMIC-IV [33] under a PhysioNet Credentialed Health Data Use Agreement. No patient re-identification was attempted. The system is designed for clinical decision *support*, not autonomous clinical decision-making.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **Yes.**

   Justification: The abstract and introduction state four specific contributions (Section 1), each supported by empirical evidence in Sections 5.1–5.4. SliceBench claims include bootstrap CIs; ClinicalBench claims use within-run paired comparisons (Section 4.3). Claims are scoped with "to our knowledge" qualifiers where appropriate.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **Yes.**

   Justification: Section 6 explicitly discusses limitations including small sample sizes (6 patients for SliceBench, 2 per tier), single-site data (MIMIC-IV), the B2→B3 confidence interval crossing zero, model-dependent effects, and physician pilot adjudication ($n = 30$, Section 5.5).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (correct) proof?

   Answer: **Yes.**

   Justification: Proposition 3 (assertion entropy loss) and Corollary 4 (faithfulness bound) in Appendix B state all assumptions explicitly. These are information-theoretic characterizations with all steps included.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

   Answer: **Yes.**

   Justification: Appendix P reports exact model versions (MedGemma 27B, Claude Sonnet 4.5, Claude Opus 4.6), temperature settings (0), bootstrap parameters ($n = 2{,}000$, seed 42, BCa method), and data source (MIMIC-IV v1.5.0 under PhysioNet DUA). Section 4 describes question generation and evaluation protocols.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

   Answer: **Yes.**

   Justification: A standalone reproducibility package is included as supplementary material (`epikg-benchmark/`) containing all 400 questions, raw predictions for six condition$\times$model combinations, and the deterministic keyword evaluator. All main-text accuracy numbers are independently reproducible. MIMIC-IV source notes require separate PhysioNet credentialed access. Full system code will be released upon acceptance.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details necessary to understand the results?

   Answer: **Yes.**

   Justification: Section 4 specifies benchmark design, question counts, condition definitions, patient selection criteria, and evaluation methodology. Appendix P provides model versions and computational requirements.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly?

   Answer: **Yes.**

   Justification: Table 5 reports BCa bootstrap 95% CIs for all conditions. Per-tier results (Figure 3) are clearly labeled as point estimates without per-tier CIs due to small tier-level sample sizes ($n = 2$ patients per tier). The overall B2$\rightarrow$B3 CI crossing zero is explicitly noted.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

   Answer: **Yes.**

   Justification: Appendix P reports computational requirements: ClinicalBench Opus $\approx$22 min per condition (API), MedGemma $\approx$3.6 hours (single GPU), SliceBench $\approx$2 hours (API).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

   Answer: **Yes.**

   Justification: The study uses de-identified clinical data under an approved data use agreement (Appendix X). No patient re-identification was attempted. The system is designed for clinical decision *support*, not autonomous decision-making.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work?

    Answer: **Yes.**

Justification: Section 6 discusses the potential for improved clinical decision support (positive) and risks of over-reliance on automated systems, context dilution, and the need for physician oversight (negative). Appendix X addresses data ethics.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models?

    Answer: **Yes.**

    Justification: The system operates on de-identified data under PhysioNet DUA. It is designed as a decision support tool requiring physician oversight. Code release will include usage guidelines emphasizing the system is not validated for autonomous clinical use.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: **Yes.**

    Justification: MIMIC-IV is cited [33] and used under PhysioNet Credentialed Health Data Use Agreement. All benchmark frameworks and models are cited. OMOP CDM is an open standard.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: **Yes.**

    Justification: ClinicalBench and SliceBench are new benchmarks fully documented in Section 4, with question generation procedures, category definitions (Appendix Q), and evaluation protocols. Templates and scoring scripts will accompany the code release.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots?

    Answer: **N/A.**

    Justification: No crowdsourcing was used. Physician adjudication (Appendix N) involves expert review by co-authors, not human subjects research.

15. **IRB approvals or equivalent**

    Question: Does the paper describe potential risks and does it include the IRB approval number if applicable?

    Answer: **N/A.**

    Justification: MIMIC-IV is a de-identified public dataset; use under PhysioNet DUA does not require separate IRB approval per PhysioNet policy.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods?

Answer: **Yes.**

Justification: LLMs are core experimental components, not authoring aids. Claude Opus 4.6 is the primary answering model for ClinicalBench (Section 5.1); Claude Sonnet 4.5 answers SliceBench questions while Opus 4.6 judges (Section 5.4); MedGemma 27B provides cross-model validation (Section 5.3). Model versions, temperatures, and inference details are in Appendix P.